

Bacillus velezensis HBXN2020 alleviates *Salmonella* Typhimurium-induced colitis by improving intestinal barrier integrity and reducing inflammation

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Abstract

Bacillus velezensis is a novel species of *Bacillus* that has been widely investigated because of its broad-spectrum antimicrobial activity. However, most studies on *Bacillus velezensis* have focused on biocontrol of plant diseases, with few reports on antagonizing *Salmonella* Typhimurium infections. In this study, *Bacillus velezensis* HBXN2020 was isolated from healthy black piglets and was found to exhibit broad-spectrum antibacterial activity and robust anti-stress capabilities. Importantly, *Bacillus velezensis* HBXN2020 did not cause any adverse side effects in mice when administered at various doses (1×10^7 , 1×10^8 , and 1×10^9 CFU) for 14 days. In a *Salmonella* Typhimurium ATCC14028-induced mice colitis model, either curative or prophylactic, supplementing *Bacillus velezensis* HBXN2020 spores significantly lowered the levels of *Salmonella* Typhimurium ATCC14028 in their feces, ileum, cecum, and colon, and the disease activity index (DAI). Importantly, supplementing *Bacillus velezensis* HBXN2020 spores significantly regulated cytokine levels (TNF- α , IL-1 β , IL-6, and IL-10) and maintained the expression of tight junction proteins and mucin protein. More importantly, supplementing *Bacillus velezensis* HBXN2020 spores also significantly enhanced the homeostasis of colonic microbiota and the abundance of beneficial bacteria. Collectively, *Bacillus velezensis* HBXN2020 can alleviate bacterial colitis by enhancing intestinal homeostasis and gut barrier integrity and reducing inflammation.

eLife assessment

In this **useful** study, Wang and colleagues investigate the potential probiotic effects of *Bacillus velezensis* to prevent colitis in a mouse model. They provide **solid** evidence that *B. velezensis* limits the growth of *Salmonella typhimurium* in lab culture and in mice, together with beneficial effects on the microbiota. The work will be of interest to infectious disease researchers and those studying the microbiome.

Introduction

Salmonella Typhimurium (*S. Typhimurium*) is a major foodborne zoonotic pathogen that can cause diarrhea and colitis in humans and animals (1, 2). It has been reported that *Salmonella* infections often occur in both developing and developed countries, which poses a substantial risk to public health and causes huge economic loss (3, 4). At present, antibiotic therapies is one of the most effective strategies for *Salmonella* infections. However, numerous studies have reported that the prolonged use or misuse of antibiotics can lead to environmental pollution, an increase in multi-drug-resistant (MDR) bacteria, as well as gastrointestinal microbiota dysbiosis (5–7). Therefore, there is an emergent search for novel antimicrobial agents to combat *Salmonella* infections.

Recently, an increasing number of studies have shown that gut microbiota plays an important role in alleviating intestinal inflammation (8) and repairing the intestinal barrier (9, 10). The use of probiotics is a popular approach to modulate intestinal microbiota nowadays. Among all probiotics, *Bacillus* is one of the most popular probiotic species because of their ability to form endospores that survive gastric transit, storage and delivery conditions (11). In addition, *Bacillus* species can also produce antimicrobial metabolites against enteric pathogens and provide extracellular enzymes, thereby reducing infection risk and improving nutrient utilization (12, 13).

Bacillus velezensis (*B. velezensis*), a newly discovered species of the *Bacillus* genus, can produce an abundance of secondary metabolites with broad-spectrum antibacterial activity (14). Previous studies have reported that *B. velezensis* exhibits varying degrees of probiotic effects on plants, livestock, poultry and fish. For instance, the combined use of *B. velezensis* SQR9 mutant and FZB42 improved root colonization and the production of secondary metabolites that are beneficial to cucumber (15). *B. velezensis* isolated from yaks was shown to enhance growth performance and ameliorate blood parameters related to inflammation and immunity in mice (16). The dietary *B. velezensis* supplementation can regulate the innate immune response in the intestine of crucian carp and reduce the degree of intestinal inflammation damage induce by *A. veronii* (17). However, most studies on *B. velezensis* were mainly biocontrol of plant fungal diseases, with few reports on the treatment or prevention of *S. Typhimurium*-induced colitis.

In this study, we have screened a strain of *B. velezensis* HBXN2020 with broad-spectrum antibacterial activity from a large number of *Bacillus* strains. We aimed to investigate the effectiveness of *B. velezensis* HBXN2020 in alleviating *S. Typhimurium*-induced mouse colitis and assessed the biological safety of *B. velezensis* HBXN2020. Notably, *B. velezensis* HBXN2020 can alleviate symptoms and colon tissue damage in experimental colitis, as indicated by markers such as *S. Typhimurium* loads, TNF- α and ZO-1 levels. Moreover, *B. velezensis* HBXN2020 was also enhanced the homeostasis of colonic microbiota and the abundance of beneficial bacteria (*Lactobacillus* and *Akkermansia*). Therefore, this study provides support for the development of probiotic-based microbial products as an alternative prevention strategy for *Salmonella* infection.

Results

Genomic Characteristics

The Illumina Hiseq and PacBio platforms were used to perform whole genome sequencing of HBXN2020, yielding a total of 8,194,986 Illumina clean reads (1,237,442,886 bases) and 225,593 PacBio reads (1,807,288,759 bases). Following a quality assessment via k-mer and a GC-depth

analysis, the whole genome was assembled *de novo* using Unicycler (Version 0.4.8). The results showed that HBXN2020 has a circular chromosome of 3,929,792 bp (**Fig. 1A**). The genome of HBXN2020 was predicted to contain 3,857 genes, including 3,744 protein-coding genes (CDS), 86 tRNA genes, and 27 rRNA genes (Table S1).

A phylogenetic tree based on genome-wide data from all 14 *Bacillus* strains demonstrated that the HBXN2020 belongs to the *B. velezensis* species (**Fig. 1B**). To further understand the classification status of HBXN2020, the online tool JSpeciesWS was used to calculate the average nucleotide identity (ANI) based on the BLAST (ANIb) method. As shown in **Fig. 1C**, HBXN2020 was found to be a member of the *B. velezensis* species due to the high percentage of ANIb (more than 97%).

Growth curve of *B. velezensis* HBXN2020 and its *in vitro* resistance against environmental assaults

The growth of *B. velezensis* HBXN2020 was assessed in flat-bottomed 100-well microtiter plates by measuring the values of OD₆₀₀ every hour using an automatic growth curve analyzer. The OD₆₀₀ value of *B. velezensis* HBXN2020 showed slight fluctuations within the first 2 h, followed by a logarithmic growth phase that reached a stable phase after 10 h (**Fig. 2A**). Despite environmental restrictions such as limited space and nutrients, the OD₆₀₀ value of the bacteria was not significantly reduced even after 22 h, indicating that the growth of *B. velezensis* HBXN2020 was relatively stable.

Next, we evaluated the effect of physical, chemical, and biological sterilization conditions such as high temperature, strong acidity, and enzyme preparation on the survival of *B. velezensis* HBXN2020. The survival rate of *B. velezensis* HBXN2020 showed a decreasing trend with increasing temperature. At 65°C, the survival rate of *B. velezensis* HBXN2020 was less than 1%, while the spores of *B. velezensis* HBXN2020 exceeded 90% (**Fig. 2B**). The survival rate of *B. velezensis* HBXN2020 spores was also above 80% even under high-temperature conditions (95°C). Similar results were observed following exposure to a strong acid environment (pH 2.0) (**Fig. 2C**), whereby *B. velezensis* HBXN2020 spores were found to effectively tolerate strong acidic conditions and exhibited improved survival rate. Based on these results, we hypothesized that *B. velezensis* HBXN2020 spores might also exhibit improved survival in gastrointestinal tract environments. Therefore, we further evaluated the survival rate of *B. velezensis* HBXN2020 in bile salt (0.3%, v/v), simulated gastric fluid (pepsin 1 mg mL⁻¹, pH 1.2), and simulated intestinal fluid (trypsin 1 mg mL⁻¹, pH 6.8). As shown in **Fig. 2D-F**, *B. velezensis* HBXN2020 spores demonstrated significantly improved tolerance to these simulated gastrointestinal tract environments.

Antibiotic susceptibility of *B. velezensis* HBXN2020 and bacteriostasis effect *in vitro*

To evaluate the drug resistance of *B. velezensis* HBXN2020, 19 commonly used antibiotics in clinical practice were selected. The results indicated that only polymyxin B had a relatively small inhibition zone diameter (less than 15 mm). Ampicillin, meropenem, minocycline, ofloxacin, and trimethoprim had the strongest inhibition on *B. velezensis* HBXN2020, with an inhibition zone diameter exceeding 30 mm (**Fig. 3A**). In addition, the experimental results showed that *B. velezensis* HBXN2020 was extremely sensitive to β -lactams, tetracyclines, and quinolone drugs. Then, we performed an inhibition test of *B. velezensis* HBXN2020 against seven common clinical pathogenic bacteria. The results showed that *B. velezensis* HBXN2020 had a similar inhibitory effect on standard and wild strains of *E. coli*, *Salmonella*, *S. aureus*, and *C. perfringens*, as well as wild strains of *S. suis*, *P. multocida*, and *A. pleuropneumoniae*. Except for the wild strains of *S. suis* and *A. pleuropneumoniae*, the diameter of the inhibition zone of other strains was above 15 mm (**Fig. 3B**, S3), while the size of the inhibition zone of *S. suis* and *A. pleuropneumoniae* was also more than 12 mm.

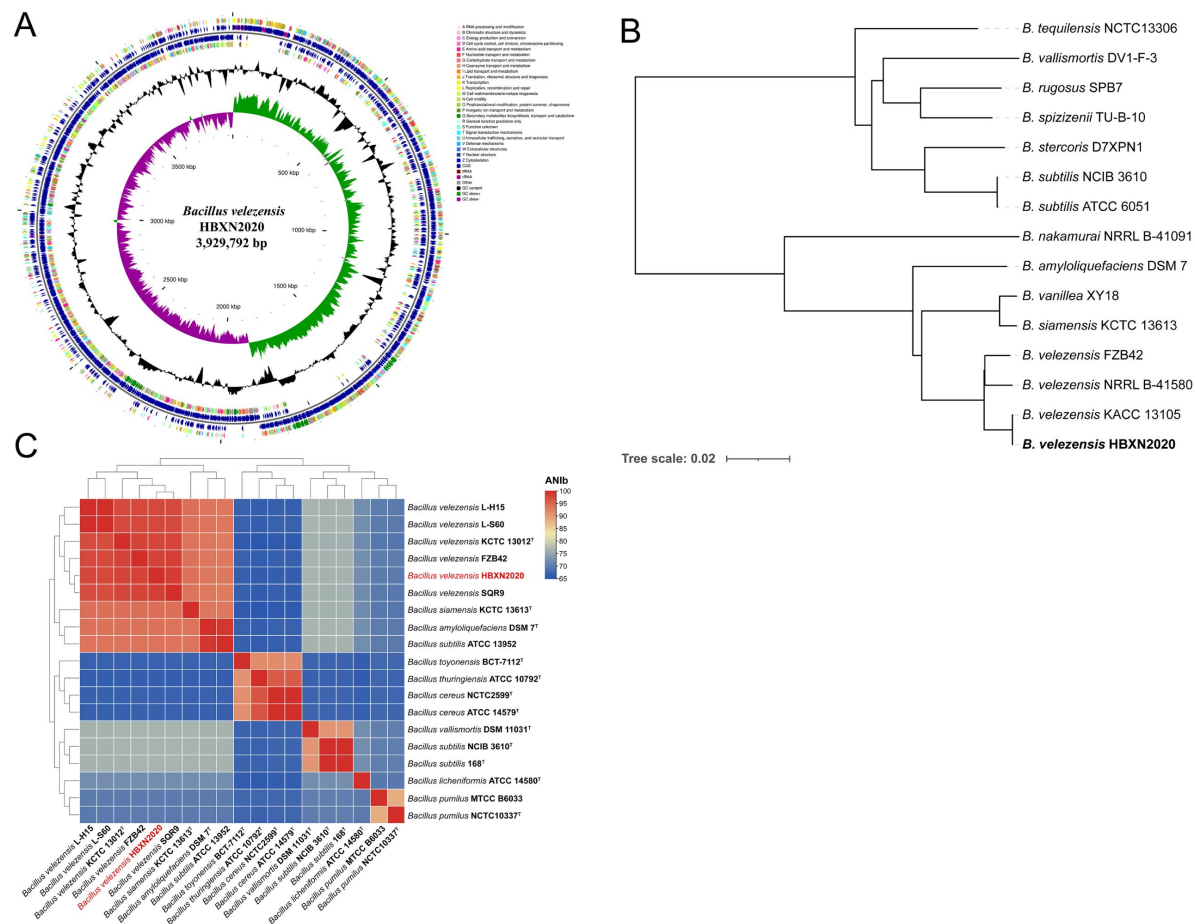


Fig. 1

Genomic characteristics and phylogenetic relationships of *B. velezensis* HBXN2020.

(A) The whole-genome map of *B. velezensis* HBXN2020 with its genomic features. The map consists of 6 circles. From the inner circle to the outer circle: (1) GC-skew, (2) GC content, (3) reverse protein-coding genes, different colors represents different COG functional classifications, (4) genes transcribed in reverse direction, (5) genes transcribed in forward direction, (6) forward protein-coding genes, different colors represents different COG functional classifications. (B) The whole genome phylogenetic tree was constructed based on genome-wide data from 14 bacillus strains. *B. velezensis* HBXN2020 are indicated in bold. (C) The heatmap based on the ANIb value of strain HBXN2020 and other *Bacillus* species. The *B. velezensis* HBXN2020 was labeled in red letters.

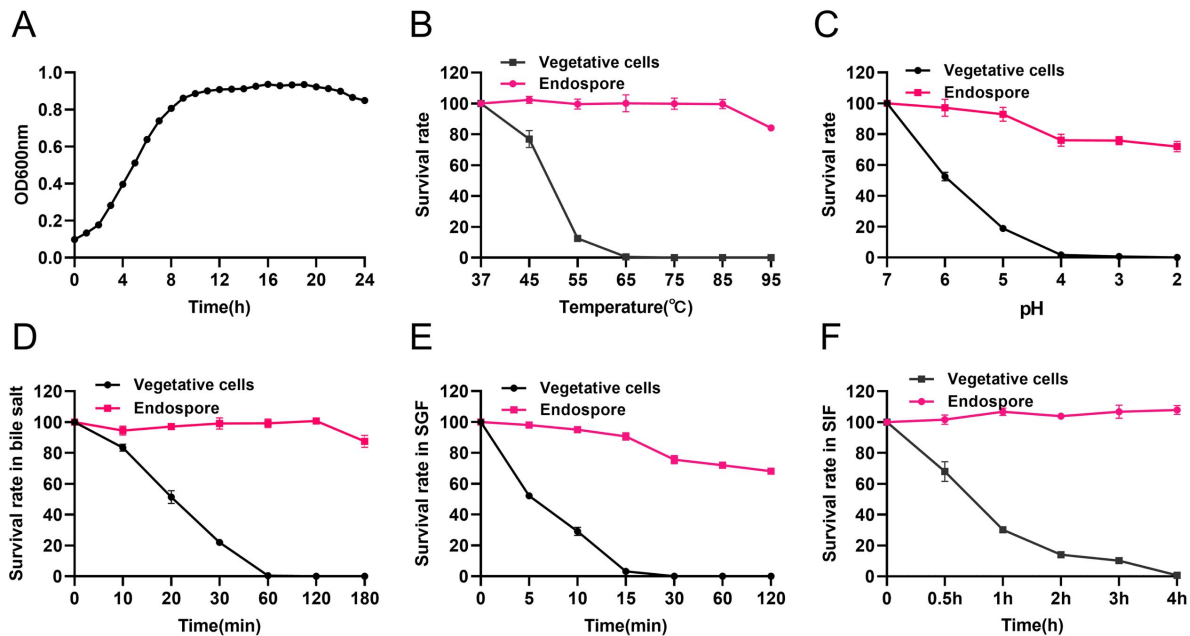


Fig. 2

Growth curve of *B. velezensis* HBXN2020 and its *in vitro* resistance against environmental assaults.

(A) Growth curves of *B. velezensis* HBXN2020 cultured in LB medium at 37°C, detection of OD₆₀₀ values at 1 h intervals in microplate reader. (B) Survival rate of endospore and vegetative cells of *B. velezensis* HBXN2020 after 30 min at different temperatures (37°C, 45°C, 55°C, 65°C, 75°C, 85°C or 95°C). Equal amounts of endospore and vegetative cells of HBXN2020 were exposed to the following: (C) acid solution (pH 2 to 7), (D) 0.3% bile salts, (E) SGF (pH 1.2) supplemented with pepsin, and (F) SIF (pH 6.8) containing trypsin at 37°C. At predetermined time points, 100 µL was taken from each sample, and tenfold serially diluted with sterile PBS (pH 7.2), and then spread on LB agar plates and cultured at 37°C for 12 h before bacterial counting. Each group was repeated three times (n = 3).

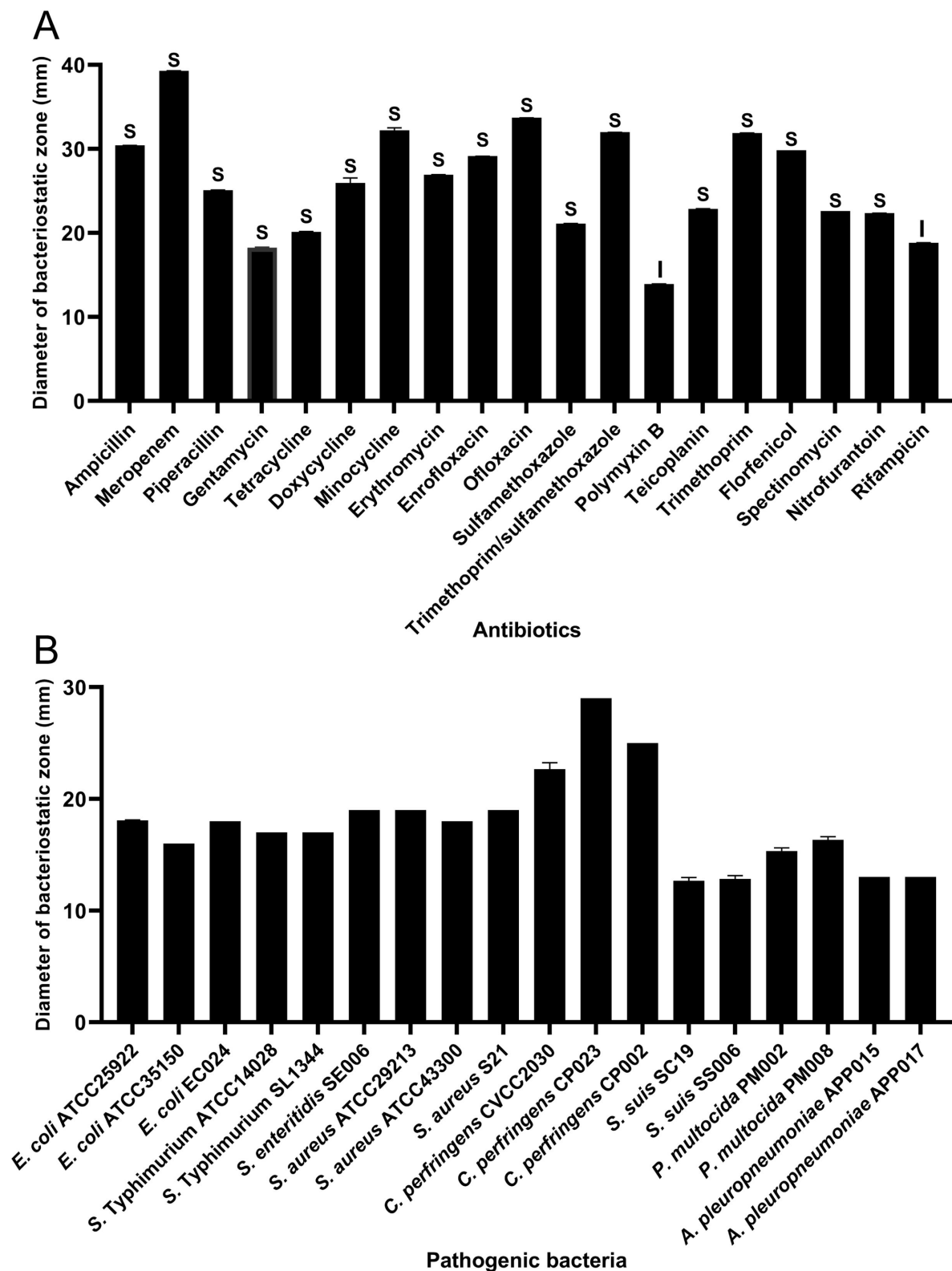


Fig. 3

Antibiotic susceptibility of *B. velezensis* HBXN2020 and bacteriostasis assay *in vitro*.

(A) The diameter of the antibacterial zone indicates the extent of sensitivity to antibiotics. (B) The diameter of the antibacterial zone indicates the extent of inhibition against pathogenic bacteria. The diameter of the antibacterial zone was measured with vernier caliper. Each group was repeated three times (n = 3). R, resistant; I, moderately sensitive; S, sensitive.

Biosafety evaluation of *B. velezensis* HBXN2020

To determine the safety of *B. velezensis* HBXN2020, we evaluated its biological safety in a mouse model. After gavage with *B. velezensis* HBXN2020 spores for two weeks, we observed no significant difference in the body weight of each group of mice (Fig. 4A). The gene expression levels of TNF- α , IL-1 β , IL-6, and IL-10 of colon from all groups of mice exhibited no remarkable changes (Fig. 4B). However, in the colon, mRNA levels of the barrier proteins ZO-1 and occludin were trending towards an increase compared with the control group (Fig. 4C). Additionally, blood routine tests and serum biochemistry tests were performed for mice in the control group and H-HBXN2020 group on the 14th day after oral administration of *B. velezensis* HBXN2020 spores multiple times. As shown in Fig. 4D and S1, the blood parameters of mice after *B. velezensis* HBXN2020 treatment, including red blood cells (RBC), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), hemoglobin (HGB), white blood cells (WBC), platelets (PLT), hematocrit (HCT), and mean corpuscular hemoglobin concentration (MCHC), were consistent with those of healthy mice. The serum biochemical parameters, including alanine aminotransferase (ALT), aspartate aminotransferase (AST), albumin (ALB), total bilirubin (TBIL), serum creatinine (CREA), and blood urea nitrogen (BUN), were also within normal limits (Fig. 4E). The corresponding histological analysis of colon tissue from mice receiving low, medium, and high doses of *B. velezensis* HBXN2020 spores (L-HBXN2020, M-HBXN2020, and H-HBXN2020 groups, respectively) is presented in Fig. 4F. The colon tissue sections of mice in the test groups showed no significant differences compared to the control group. Additionally, there were no observable differences in the major organ tissues (heart, liver, spleen, lung, and kidney) of mice treated with the high dose of *B. velezensis* HBXN2020 spores compared to healthy mice (Fig. S2). Collectively, these experimental results indicate that *B. velezensis* HBXN2020 is safe for use and does not cause any adverse side effects in mice.

Oral administration of *B. velezensis* HBXN2020 spores alleviated *S. Typhimurium*-induced colitis

To investigate whether *B. velezensis* HBXN2020 could inhibit *S. Typhimurium* ATCC14028 (STm), solid and liquid co-culture tests were conducted, and the results showed that *B. velezensis* HBXN2020 inhibited the growth of STm in a dose-dependent manner (Fig. 5A, S4A-B, S4D). Next, the therapeutic potential of *B. velezensis* HBXN2020 was evaluated in an STm-induced mouse colitis model. At days 1, 3, and 5 after STm infection, mice in the STm + HBXN2020 group were orally administered *B. velezensis* HBXN2020 spores by gavage (Fig. 5B), while the control group and STm + PBS group were treated with sterile PBS. Also, the number of excreted STm in feces was counted daily at the designated time points. As shown in Fig. 5C, there was a significant and continuous reduction in the number of STm in feces following treatment with *B. velezensis* HBXN2020 spores, while the number of *B. velezensis* HBXN2020 viable bacteria in feces is also gradually decreasing (Fig. S5A). The number of STm in mouse feces was reduced by 10 to 50 times compared to the STm + PBS group from day 2 to 7 post-treatment. *B. velezensis* HBXN2020 spore treatment also resulted in a 5 to 10-fold reduction in the number of STm in the ileum, cecum, and colon, respectively (Fig. 5D-E, S5B). The increased inhibition of STm after *B. velezensis* HBXN2020 spore treatment also resulted in less weight loss and a marked reduction ($P < 0.001$) in DAI scores in STm + HBXN2020 group mice (Fig. 5F-G). In addition, we also measured the colon length of the mice and found that the STm + PBS group had a significantly shorter colon length ($P < 0.001$) than the control group. However, compared to the STm + PBS group, the STm + HBXN2020 group exhibited a longer colon ($P < 0.05$) (Fig. 5H-I).

To further assess the impact of *B. velezensis* HBXN2020 on the intestinal tract, we histologically analyzed their colonic tissues. The results showed that mice treated with *B. velezensis* HBXN2020 spores exhibited a lower histology score (total scoring) ($P < 0.05$) (Fig. S5C) than mice not treated with the spores and were similar to that of healthy mice (control group). The colon tissue of mice not treated with *B. velezensis* HBXN2020 spores displayed higher histological scores, with

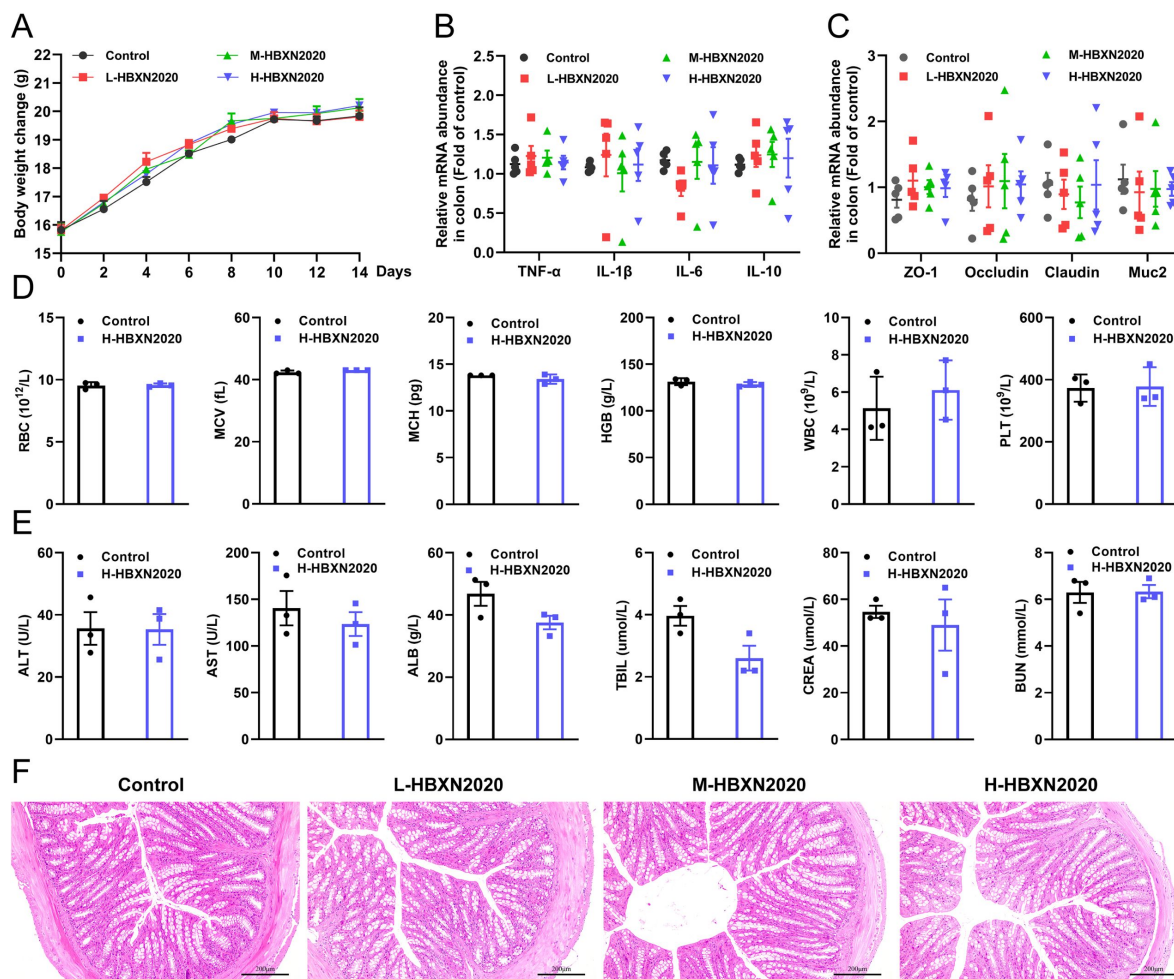


Fig. 4

***In vivo* safety evaluation of *B. velezensis* HBXN2020 in a mouse model.**

(A) Body weights changes of mice during gavage with *B. velezensis* HBXN2020 spores. Mice were treated with sterile PBS (Control group) or low-dose (L-HBXN2020 group), medium dose (M-HBXN2020 group), and high-dose (H-HBXN2020 group) of *B. velezensis* HBXN2020 spores. Weighing and gavage were performed once every two days during the experimental period (15 days). Data were shown as mean values $\pm$ SEM ($n = 5$). (B) The relative gene expression levels of inflammatory cytokines in the colon of mice measured by RT-qPCR. Data were shown as mean values $\pm$ SEM ($n = 5$). (C) The relative gene expression levels of barrier protein ZO-1, occludin, claudin and Muc2 in the colon of mice measured by RT-qPCR. Data were shown as mean values $\pm$ SEM ($n = 5$). (D) Major blood routine parameters and (E) serum biochemical parameters of mice in the control group and H-HBXN2020 group. Data were shown as mean values $\pm$ SEM ($n = 3$). (F) H&E stained colon sections in the different groups. Scale bar: 200 μ m.

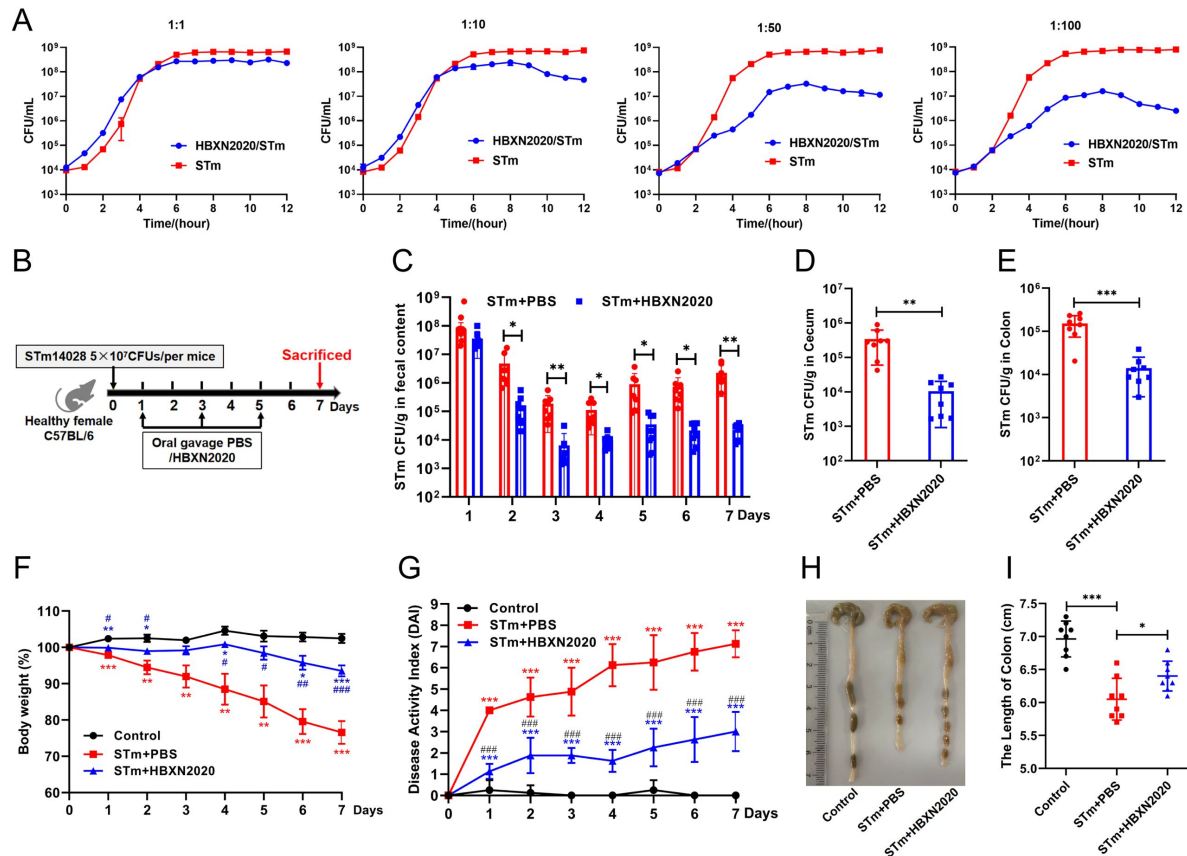


Fig. 5

Oral *B. velezensis* HBXN2020 spores alleviated *S. Typhimurium*-induced experimental mouse colitis.

(A) *In vitro* Bacterial competition between STm and *B. velezensis* HBXN2020. STm were co-incubated with *B. velezensis* HBXN2020 at various ratios at 37°C with shaking. The growth of STm was reflected by bacterial counting per hour. (B) Experimental design for treatment in this study. Orally administrated with either PBS, or *B. velezensis* HBXN2020 spores by gavage at days 1, 3, and 5 after STm infection (5×10^7 CFU/mouse), respectively. All mice were euthanized at day 7 after STm infection. (C) Bacterial count of STm in mouse feces. Fecal samples were collected per day after STm infection and resuspended in sterile PBS (0.1 g of fecal resuspended in 1 mL of sterile PBS). One hundred microliters of each sample performed a serial of 10-fold dilutions and spread on selective agar plates ($50 \mu\text{g mL}^{-1}$ kanamycin) and incubated at 37°C for 12 h before bacterial counting. The bacterial loads of STm in (D) cecum, and (E) colon. The cecum and colon were harvested and then homogenized. One hundred microliters of each sample performed a serial of 10-fold dilutions and spread on selective agar plates ($50 \mu\text{g mL}^{-1}$ kanamycin) and incubated at 37°C for 12 h before bacterial counting. Statistical significance was evaluated using Student's t-test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$). (F) Daily body weight changes and (G) daily disease activity index (DAI) scores of mice with different treatment groups. Data were shown as mean values $\pm$ SEM ($n = 8$). Statistical significance was evaluated using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test. *, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$ relative to Control group; #, $P < 0.05$, ##, $P < 0.01$, ###, $P < 0.001$ relative to STm + HBXN2020 group. (H) Colonic tissue images. (I) The length of the colon from per group ($n = 8$). Statistical significance was evaluated using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

significant crypt deformation, severe mucosal damage, and obvious inflammatory cell infiltration. These findings suggest that *B. velezensis* HBXN2020 can alleviate colon tissue damage. Moreover, we measured the mRNA levels of inflammatory cytokines, including tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), interleukin-6 (IL-6), and interleukin-10 (IL-10) in colon tissue using RT-qPCR analysis. As shown in **Fig. 6B-D**, the mRNA levels of TNF- α ($P < 0.01$), IL-1 β ($P < 0.05$), and IL-6 ($P < 0.05$) in the colon tissue of STm-infected mice were significantly upregulated, indicating the successful establishment of the STm-induced colitis model. As expected, treatment with *B. velezensis* HBXN2020 spores led to a significant reduction in the mRNA levels of these pro-inflammatory cytokines, while the mRNA levels of the anti-inflammatory cytokine IL-10 were significantly increased ($P < 0.01$) (**Fig. 6E**), suggesting that *B. velezensis* HBXN2020 spores can alleviate the inflammatory reaction of the colon. Moreover, the mRNA expression levels of colonic epithelial barrier proteins, including ZO-1, occludin, claudin, and Muc2, were also measured by RT-qPCR. Compared with the PBS-treated group, the *B. velezensis* HBXN2020 spores treatment group exhibited significantly upregulated mRNA expression levels of ZO-1 ($P < 0.05$), occludin ($P < 0.05$), claudin ($P < 0.05$), and Muc2 ($P < 0.01$) in the colon tissue (**Fig. 6F-I**), further demonstrating the ability of *B. velezensis* HBXN2020 to reduce damage to the colon epithelial barrier.

Next, we further explored the effect of HBXN2020 spores on the intestinal microbiota composition of STm-treated mice via 16S rRNA gene sequencing. The alpha diversity of colonic microbiota was evaluated using Sobs, Chao1, Shannon, and Simpson indices (OTU levels). Compared with the control and HBXN2020 group, the Sobs, Chao1, and Shannon indices of the STm + PBS group clearly decreased (**Fig. 7A-C**), while the Simpson index significantly increased ($P < 0.05$, **Fig. 7D**). Principle component analysis (PCA) based on Bray–Curtis distance showed that compared with the STm + PBS group, the gut microbiota composition of the STm + HBXN2020 group and the control group were more closer together ($R = 0.3988$, $P = 0.001$, **Fig. 7E**). We analyzed the significance of differences among groups at different classification levels (**Fig. 7F-G**), and the results revealed that the composition of the gut microbiota changed markedly after STm-induced mice. At the phylum level, the STm + PBS group increased the abundance of *Verrucomicrobiota* and *Proteobacteria* and decreased the abundance of *Firmicutes*, *Patescibacteria*, *Desulfobacterota*, and *Actinobacteriota* compared to the Control group (**Fig. 7F**). The HBXN2020 group increased the abundance of *Bacteroidota* and *Firmicutes* and decreased the abundance of *Verrucomicrobiota* compared with the Control group. On the contrary, the relative abundance of *Firmicutes*, *Patescibacteria*, *Desulfobacterota*, and *Actinobacteriota* in the STm + HBXN2020 group was significantly higher than that in the STm + PBS group. At the genus level, infection with *S. Typhimurium* ATCC14028 in the STm + PBS group dramatically reduced the relative abundance of *norank_f_Muribaculaceae* and *Lactobacillus* (**Fig. 7H-I**) and enhanced the abundance of *Bacteroides*, *Escherichia-Shigella*, *Salmonella*, *Alistipes*, and *Enterococcus* compared to the Control group (**Fig. 7J-K**, S6B). The HBXN2020 group increased the abundance of *norank_f_Muribaculaceae* and *Alloprevotella* compared to the Control group. In contrast, STm + HBXN2020 group significantly improved the relative abundance of *norank_f_Muribaculaceae* and *Lactobacillus* compared with the STm + PBS group ($P < 0.01$, $P < 0.001$, **Fig. 7H-I**). In addition, LEfSe analysis was also performed to identify specialized microbial communities between the STm + HBXN2020 group and the STm + PBS group, and found that there was a clear difference in gut microbiota composition between them. The microbiota that played a major role in the STm + PBS group were harmful *Enterobacterales* (**Fig. 7L**), while the STm + HBXN2020 group was *Candidatus_Saccharimonas* (**Fig. 7L**).

Prophylactic *B. velezensis* HBXN2020 spores alleviated *S. Typhimurium*-induced colitis

Based on the improved therapeutic efficacy of *B. velezensis* HBXN2020 in the treatment of STm-induced colitis in mice, we further explored its potential for disease prevention by evaluating pre-treatment. Mice in the PBS + STm group and HBXN2020 + STm group were given the same amount

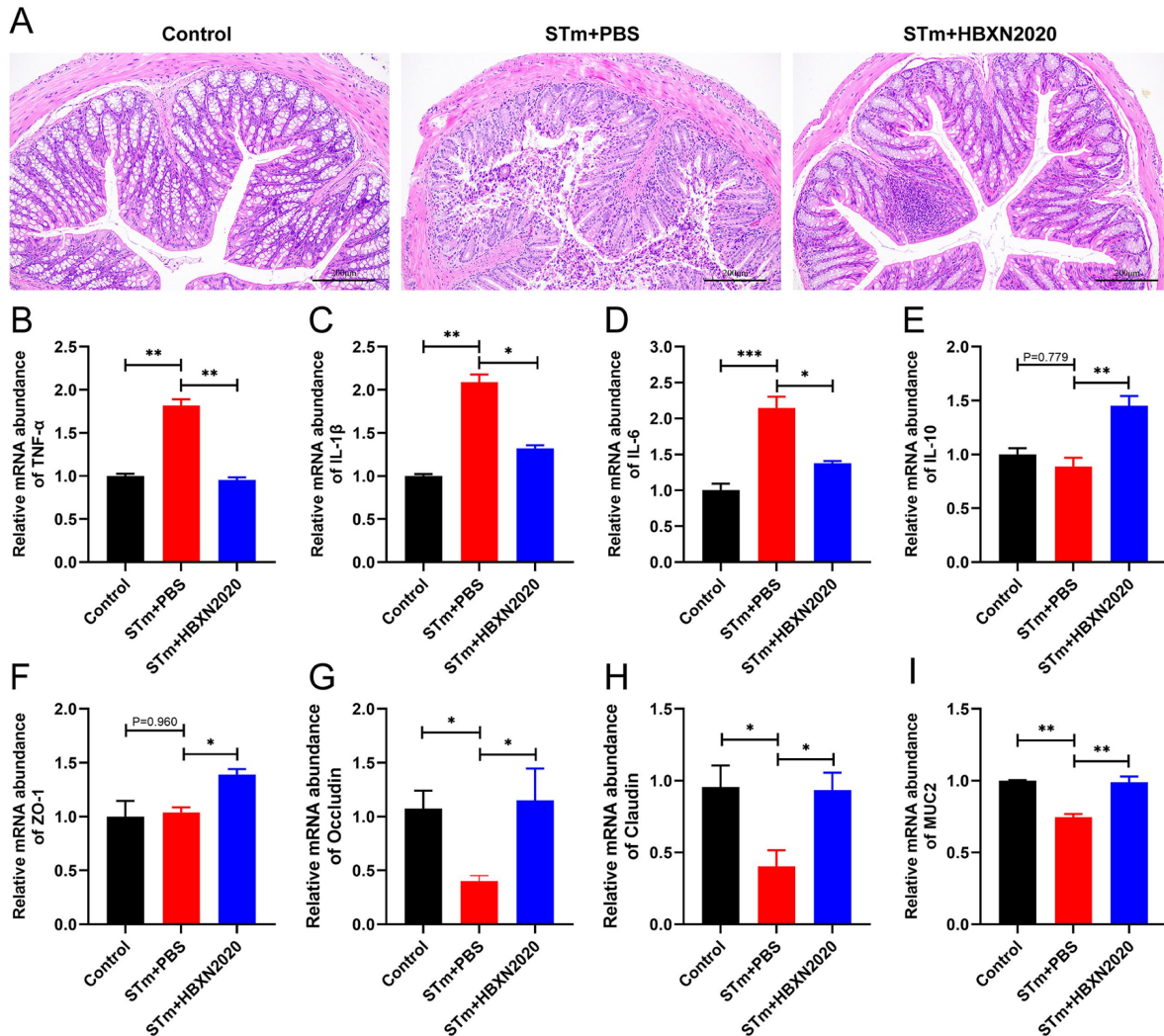


Fig. 6

Oral *B. velezensis* HBXN2020 spores attenuated colonic damage and inflammatory reaction.

(A) H&E stained colon tissue sections. Scale bar: 200 μ m. The relative gene expression levels of TNF- α (B), IL-1 β (C), IL-6 (D), and IL-10 (E) were detected by RT-qPCR. Data were shown as mean values $\pm$ SEM (n = 6). The relative gene expression levels of ZO-1 (F), occludin (G), claudin (H), and Muc2 (I) in colon tissue were detected by RT-qPCR. Data were shown as mean values $\pm$ SEM (n = 6). Statistical significance was evaluated using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

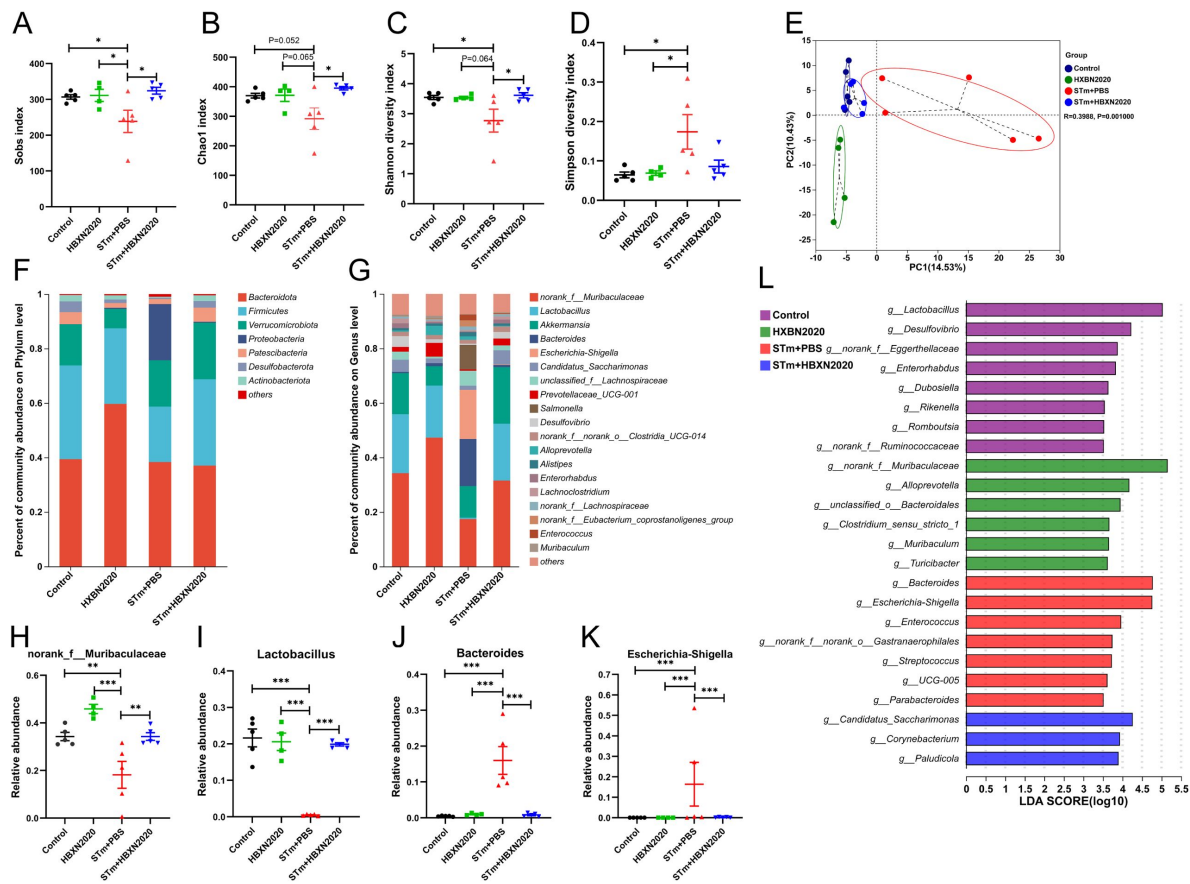


Fig. 7

Oral *B. velezensis* HBXN2020 spores regulated the composition of intestinal microbiota.

(A to D) The α -diversity of the gut microbiota, determined by the (A) Sobs, (B) Chao1, (C) Shannon, and (D) Simpson diversity index. Data were shown as mean values $\pm$ SEM ($n = 5$). (E) The PCA plot showed the β -diversity of the gut microbiota based on Bray-Curtis distance at the OTU level. (F and G) The relative abundance of colonic microbiota at the phylum (F) and genus (G) levels. (H-K) Relative abundance of selected taxa (H) *norank_f_Muribaculaceae*, (I) *Lactobacillus*, (J) *Bacteroides*, and (K) *Escherichia-Shigella*. Data were shown as mean values $\pm$ SEM ($n = 5$). (L) Analysis of differences in the microbial communities by LefSe (linear discriminant analysis (LDA) score > 3.5) among different groups. Significance was evaluated by the Kruskal-Wallis test or ANOVA with Tukey's multiple comparisons test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

of sterile PBS and *B. velezensis* HBXN2020 spores by gavage one week in advance. As shown in **Fig. 8A**, compared with the PBS + STm group, oral administration of *B. velezensis* HBXN2020 spores as a preventive measure remarkably alleviated STm-induced colitis, including weight loss of mice ($P < 0.01$) (**Fig. 8B**), a significant reduction in DAI ($P < 0.001$) (the comprehensive score of weight loss, stool consistency, and blood in the feces, **Fig. 8C**), and the prevention of colon length shortening (**Fig. 8D-E**). However, the number of *B. velezensis* HBXN2020 viable bacteria in feces is also gradually decreasing with the time prolonging (**Fig. S7A**). Furthermore, compared with the PBS + STm group, oral administration of *B. velezensis* HBXN2020 spores not only reduced the number of STm in mouse feces (**Fig. 8F**) but also decreased STm colonization in the ileum, cecum, and colon (**Fig. 8G-H**, **S7B**).

Histological analysis further revealed that oral administration of *B. velezensis* HBXN2020 significantly reduced the infiltration of inflammatory cells, mucosal damage and the overall histological score of colon tissue (**Fig. 9A**, **S7C**). Additionally, the mRNA expression levels of pro-inflammatory cytokines, including TNF- α ($P < 0.05$), IL-1 β ($P < 0.05$) and IL-6 ($P < 0.01$), in the colon tissue of mice pretreated with *B. velezensis* HBXN2020 spores were significantly decreased (**Fig. 9B-D**), while the mRNA levels of anti-inflammatory cytokine (IL-10) were significantly higher ($P < 0.05$) (**Fig. 9E**) than mice pretreated with PBS. In addition, compared to the group pretreated with PBS, the group pretreated with *B. velezensis* HBXN2020 spores showed a significant increase in mRNA expression levels of ZO-1 ($P < 0.001$), occludin ($P < 0.05$), and claudin ($P < 0.05$) in the colon tissue of mice (**Fig. 9F-H**). While there was no significant difference in Muc2 expression ($p=0.252$) (**Fig. 9I**), the results were consistent with those of the therapeutic experiments. Overall, these findings suggest that *B. velezensis* HBXN2020 can effectively alleviate the symptoms of STm-induced colitis and the associated damage to colon tissue.

Next, we examined the impact of prophylactic *B. velezensis* HBXN2020 on the intestinal microbiota composition of STm-treated mice. Sobs, Chao1 index and Shannon index have changed to some extent between groups, but differences were not significant (**Fig. 10A-C**). Compared with the Control group, the Simpson index significantly decreased in the PBS+STm group ($P < 0.05$, **Fig. 10D**). However, compared to the PBS+STm group, Simpson index in the HBXN2020+STm group did not change (**Fig. 10D**). PCA analysis showed a significant separation between control and PBS+STm groups, while the HBXN2020+STm group, HBXN2020 group, and the control group were clustered together (**Fig. 10E**). Next, we analyzed microbial community composition at the phylum and genus levels (**Fig. 10F-G**). At the phylum level, the PBS + STm group increased the abundance of *Bacteroidota*, *Proteobacteria*, *Actinobacteriota*, and *Deferribacterota* and reduced the abundance of *Verrucomicrobiota*, *Firmicutes*, *Patescibacteria*, and *Desulfobacterota* compared with the Control group (**Fig. 10F**, **S8A**). The HBXN2020 group increased the abundance of *Bacteroidota* compared to the control group. In contrast, compared to the PBS+STm group, the abundance of *Firmicutes* in the HBXN2020+STm group was significantly increased ($P < 0.05$). At the genus level, a depletion of *Lactobacillus* and *Akkermansia* (**Fig. 10H-I**) and an enrichment of *Alistipes*, *Bacteroides*, *Escherichia-Shigella*, *Alloprevotella*, *Helicobacter*, *Enterococcus*, and *Salmonella* were observed in PBS + STm group compared with the control (**Fig. 10J-M**, **S8B**). The enrichment with *norank_f_Muribaculaceae* and *Alloprevotella* were observed in HBXN2020 group compared to the control. In contrast, an enrichment of *Lactobacillus*, *Akkermansia*, and *Lachnospiraceae_NK4A136_group* and a sharp decrease of *Alistipes*, *Bacteroides*, *Escherichia-Shigella*, and *Enterococcus* were observed in HBXN2020 + STm group compared to the PBS + STm group. LEfSe analysis showed that the abundance of *Lactobacillus* in the intestinal microbiota of mice was significantly increased after *B. velezensis* HBXN2020 treatment, while the abundance of harmful *Enterobacterales* in the untreated group was clearly increased (**Fig. 10N**).

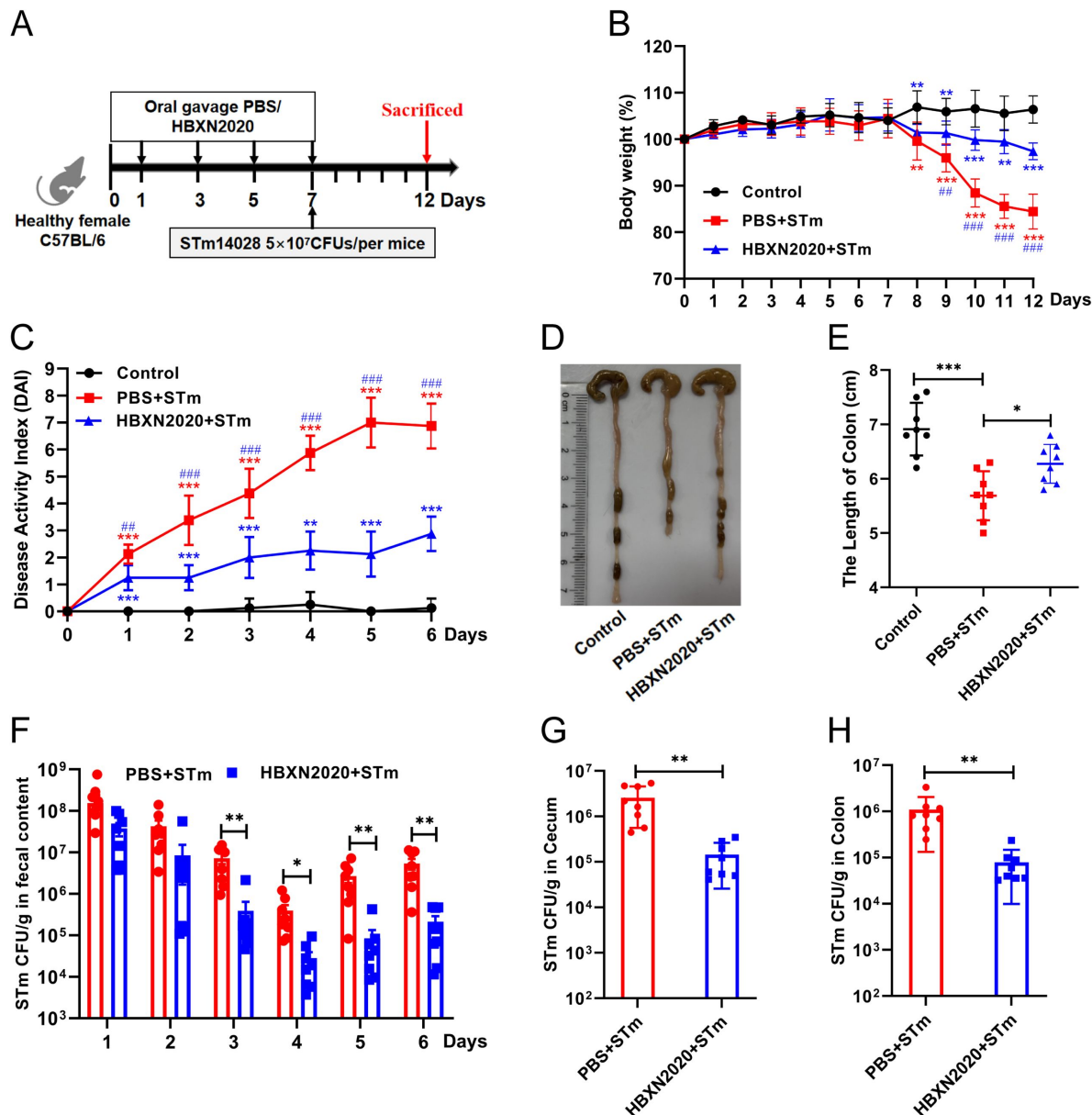


Fig. 8

Prophylactic *B. velezensis* HBXN2020 spores attenuated the symptoms of *S. Typhimurium*-induced mouse colitis.

(A) Experimental design for treatment in this study. At days 1, 3, 5, and 7, each mouse in the HBXN2020 + STm group and PBS + STm group were received 200 μ L (1×10^8 CFU/mouse) of *B. velezensis* HBXN2020 spores or sterile PBS by gavage. Then, mice in PBS + STm group and HBXN2020 + STm group were orally inoculated with 200 μ L (5×10^7 CFU/mouse) of STm on day 7. On day 12, all mice were euthanized. (B) Daily body weight changes and (C) daily disease activity index (DAI) scores of mice with different groups following STm treatment. Data were shown as mean values $\pm$ SEM ($n = 8$). Statistical significance was evaluated using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test. *, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$ relative to Control group; #, $P < 0.05$, ##, $P < 0.01$, ###, $P < 0.001$ relative to HBXN2020 + STm group. (D) Colonic tissue images. (E) The length of the colon from per group ($n = 8$). (F) Bacterial count of STm in mouse feces. Fecal samples were collected every day after STm infection and resuspended in sterile PBS (0.1 g of fecal resuspended in 1 mL of sterile PBS). One hundred microliters of each sample performed a serial of 10-fold dilutions and spread on selective agar plates ($50 \mu\text{g mL}^{-1}$ kanamycin) and incubated at 37°C for 12 h before bacterial counting. The bacterial loads of STm in (G) cecum and (H) colon. The cecum and colon were harvested and then homogenized. One hundred microliters of each sample performed a serial of 10-fold dilutions and spread on selective agar plates ($50 \mu\text{g mL}^{-1}$ kanamycin) and incubated at 37°C for 12 h before bacterial counting. Statistical significance was evaluated using Student's t-test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

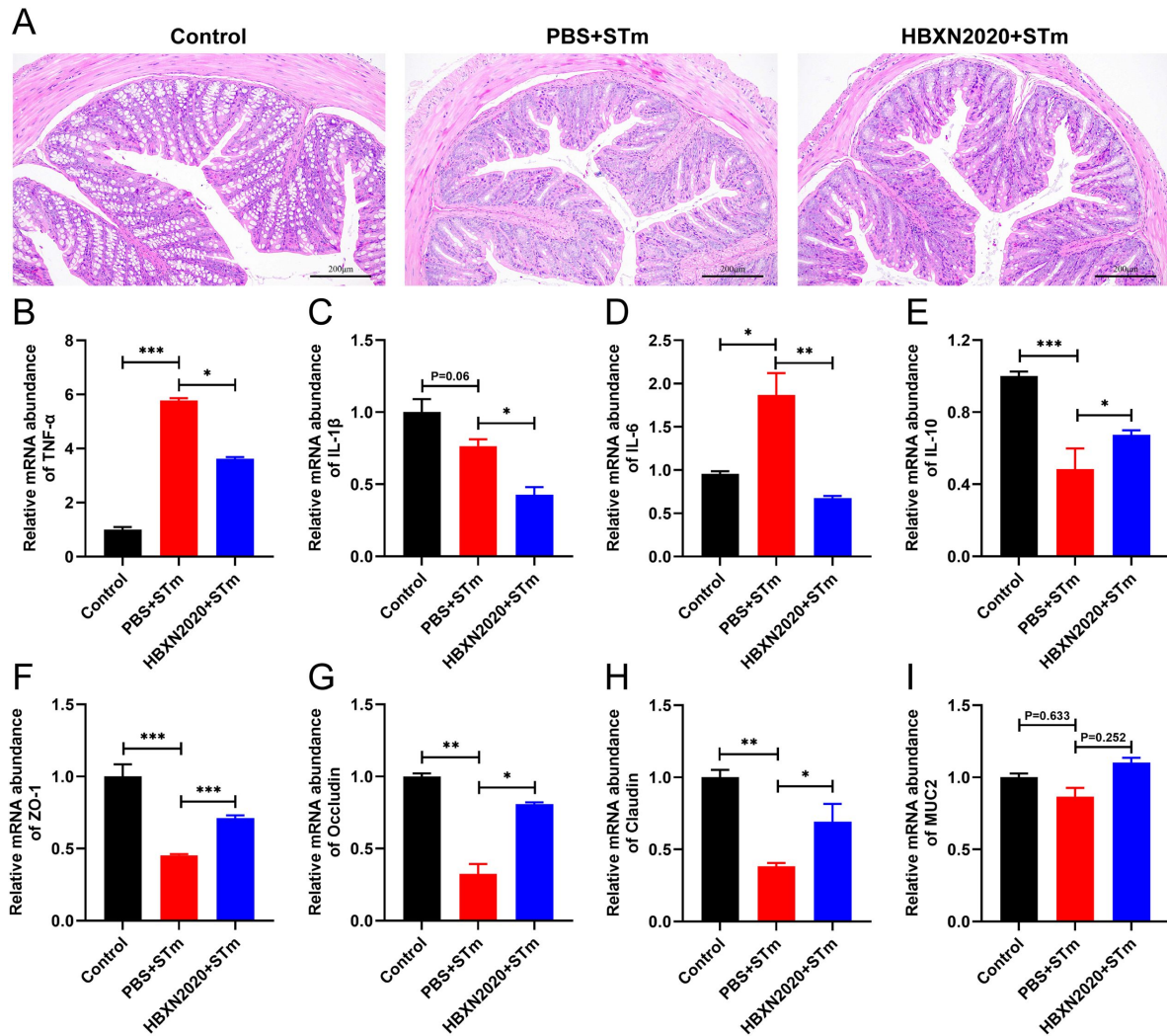


Fig. 9

Prophylactic *B. velezensis* HBXN2020 spores attenuated colonic damage and inflammatory reaction.

(A) H&E stained colon tissue sections. Scale bar: 200 μ m. The relative gene expression levels of TNF- α (B), IL-1 β (C), IL-6 (D), and IL-10 (E) were detected by RT-qPCR. Data were shown as mean values $\pm$ SEM (n = 6). The relative gene expression levels of ZO-1 (F), occludin (G), claudin (H), and Muc2 (I) in colon tissue were detected by RT-qPCR. Data were shown as mean values $\pm$ SEM (n = 6). Statistical significance was evaluated using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

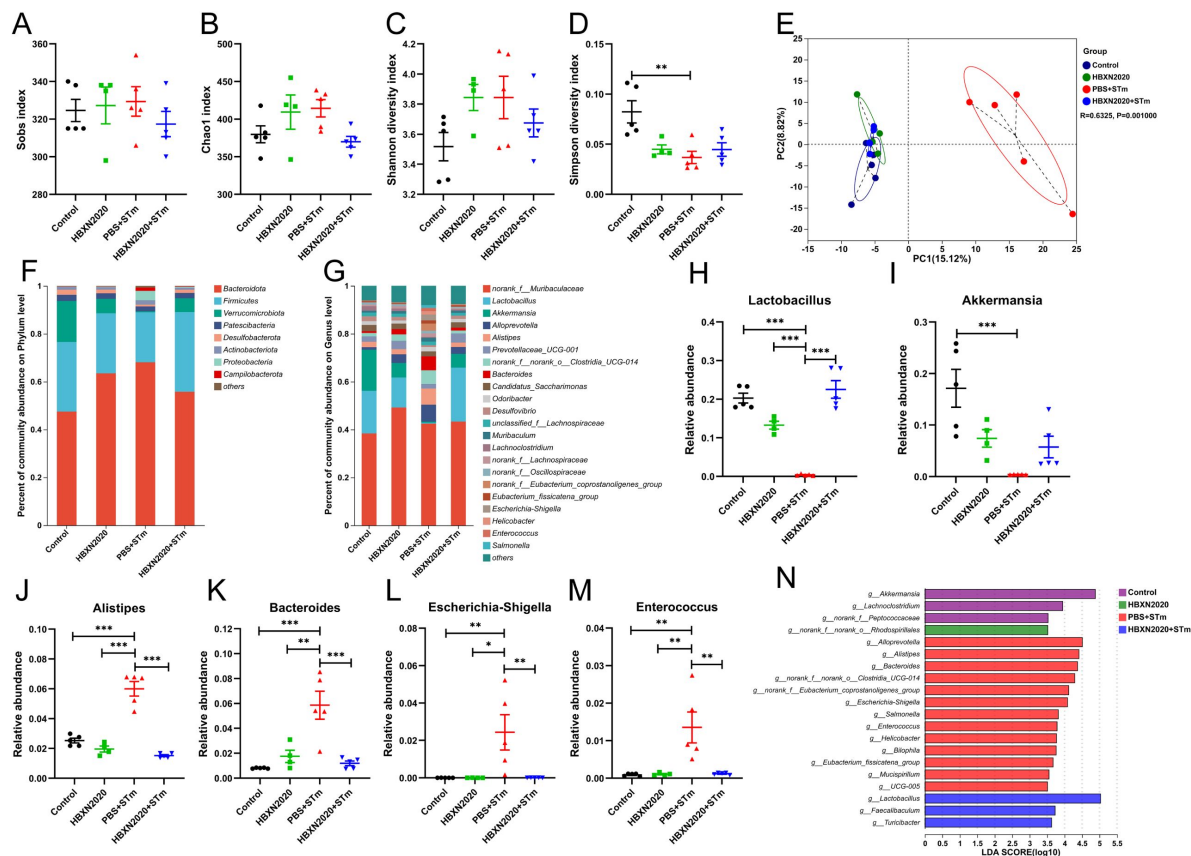


Fig. 10

Prophyllactic *B. velezensis* HBXN2020 spores regulated the composition of gut microbiota.

(A to D) Alpha diversity of the intestinal microbiota. Data were shown as mean values $\pm$ SEM ($n = 5$). (E) The PCA plot showed the β -diversity among different microbial community groups based on Bray-Curtis distance at the OTU level. (F and G) The relative abundance of colonic microbiota at the phylum (F) and genus (G) levels. (H-M) Relative abundance of selected taxa (H) *Lactobacillus*, (I) *Akkermansia*, (J) *Alistipes*, (K) *Bacteroides*, (L) *Escherichia-Shigella*, and (M) *Enterococcus*. Data were shown as mean values $\pm$ SEM ($n = 5$). (N) Analysis of differences in the microbial taxa by LeFSe (LDA score > 3.5) in different groups. Significance was evaluated by the Kruskal-Wallis test or ANOVA with Tukey's multiple comparisons test (*, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$).

Discussion

S. Typhimurium is an important intestinal pathogen that can cause invasive intestinal diseases such as bacterial colitis (18, 19). Although antibiotics are commonly used to treat bacterial-induced colitis, their overuse has significantly increased bacterial resistance in recent years, resulting in serious environmental pollution. A recent study showed that antibiotic-resistance genes in probiotics could be transmitted to the intestinal microbiota, which may threaten human health (20). Therefore, the source of probiotics is particularly important. Black pigs are an excellent domestic breed in China, with strong environmental adaptability and disease resistance (21). In this study, *B. velezensis* HBXN2020 was isolated from the free-range feces of black piglets in a mountain village in Xianning City (Hubei, China) and exhibited excellent antibacterial activity. Previous studies have demonstrated that the diffcicidins isolated from *B. subtilis* can inhibit bacterial protein synthesis, giving it broad-spectrum antibacterial activity against *Gram positive* and *Gram negative* bacteria. Therefore, we speculate that *B. velezensis* HBXN2020 has the ability to synthesize Difficidin or its derivatives (22). Otherwise, *in vitro* tolerance assays showed that *B. velezensis* HBXN2020 spores have good tolerance to high temperature, strong acids, bile salts, and simulated gastric and intestinal fluid, which was similar to previous research results (23). Furthermore, antibiotic susceptibility tests showed that *B. velezensis* HBXN2020 does not exhibit antibiotic resistance.

Probiotics are unique in that they are alive when consumed, and unlike food or drug, they have the potential to produce toxins or infections in the body (24, 25). Therefore, assessing the biosafety of probiotics is crucial. In this study, the safety of *B. velezensis* HBXN2020 was evaluated by measuring the body weight, intestinal barrier proteins, and inflammatory cytokines of the experimented mice, as well as conducting routine blood and biochemical tests. The results showed that the expression levels of ZO-1 and occludin in the colon of mice increased after treatment with different doses of *B. velezensis* HBXN2020 spores. It has been reported that ZO-1, occludin and claudin are the three most important tight junction proteins in intercellular connections, which play a crucial role in maintaining the intestinal epithelial barrier (26, 27). Therefore, upregulating the expression levels of ZO-1 and Occludin might enhance intestinal barrier function.

Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) are important aminotransferases in animals and are considered key parameters for evaluating liver function injury (28, 29). Under normal circumstances, the levels of ALT and AST are in dynamic balance without notable changes. However, when the liver is damaged or becomes dysfunctional, the levels of aminotransferases (ALT and AST) increase significantly (30). Previous studies have shown that CCL4-induced liver injury strongly increases plasma ALT/AST levels (31). Zhang et al. reported that the addition of *Bacillus subtilis* to chicken diets significantly decreased the serum levels of ALT and AST (32). In this present study, the levels of ALT and AST in the probiotic-treated group were slightly lower than those in the control group, indicating that *B. velezensis* HBXN2020 had no negative effect on liver health in mice. Furthermore, routine blood tests found that the blood indices of the *B. velezensis* HBXN2020 treatment group were within the normal reference range, similar to that of the control group.

To further evaluate the potential of *B. velezensis* HBXN2020 spores in alleviating *S. Typhimurium*-induced colitis, mice were challenged with *S. Typhimurium* ATCC14028 and then treated with *B. velezensis* HBXN2020 spores via oral administration. Our experimental results demonstrated that oral treatment with *B. velezensis* HBXN2020 spores could alleviate *S. Typhimurium*-induced colitis, as evidenced by decreased weight loss, DAI, and histological damage, which is similar with prior studies (33, 34). Previous research showed that macrophages are the first line of host defense against bacterial infection. They release pro-inflammatory cytokines, which are critical in initiating adaptive immune responses (35, 36). Nevertheless, pro-inflammatory cytokines have immunological properties that can be beneficial for the host to resist the invasion of bacteria and other microbes in the surrounding environment (37). For instance, pre-treatment with

recombinant murine TNF- α was shown to protect mice against lethal bacterial (*E. coli*) infection (38 [↗](#)). PJ-34 exerted protective effects on intestinal epithelial cells against invasive *Salmonella* infection by upregulating IL-6 expression through the ERK and NF- κ B signal pathways (39 [↗](#)). However, some studies have shown that excessive immune cell activation can lead to tissue damage, multi-organ malfunction, systemic or chronic inflammation, and autoimmune disorders (40 [↗](#), 41 [↗](#)). IL-10 is a recognized anti-inflammatory mediator that plays a crucial role in maintaining intestinal microbe-immune homeostasis, regulating the release of inflammatory mediators, and inhibiting pro-inflammatory responses of innate and adaptive immunity (42 [↗](#)–44 [↗](#)). For instance, previous studies have shown that *Clostridium butyrate* can induce the production of IL-10 in the intestine, thereby alleviating experimental colitis in mice (45 [↗](#)). Similar results were observed in our study, where oral administration of *B. velezensis* HBXN2020 spores reduced the expression levels of TNF- α , IL-1 β and IL-6 while increasing the levels of IL-10 in the colon of mice. Furthermore, oral administration of *B. velezensis* HBXN2020 spores improved the functional damage and integrity of the intestinal barrier caused by *S. Typhimurium* infection, as evidenced by the upregulated expression levels of intestinal barrier proteins and reduced histological damage.

Subsequently, we further analyzed the colon microbiome and it was noteworthy that oral *B. velezensis* HBXN2020 markedly increased alpha diversity of the colonic microbiota. As reported in prior studies, elevated alpha diversity associates with enhanced anti-interference ability (46 [↗](#)). PCA analysis Results showed that oral administration of *B. velezensis* HBXN2020 significantly affected the structure of the intestinal microbiota. Further analysis at the phylum/genus level revealed that *B. velezensis* HBXN2020 treatment reduced the increase of *Bacteroides*, *Escherichia-Shigella*, and *Salmonella* (harmful to intestinal homeostasis) in colitis (47 [↗](#)), but enhanced the relative abundance of *Lactobacillus* (known for enhancing the epithelial barrier by increasing mucus secretion and upregulating the expression of tight junction proteins such as claudin-1, occludin, and ZO-1) (48 [↗](#)) and *Akkermansia* (known for modulating intestinal immune response by producing short chain fatty acids and reducing the secretion of pro-inflammatory cytokines) (49 [↗](#)). *Bacteroides* is generally ‘friendly’ commensals in the intestine, but they can enter normal tissues through the intestinal mucosa when intestinal barrier function becomes compromised (50 [↗](#)), leading to inflammation, diarrhea, and abdominal abscess (51 [↗](#)). *Escherichia-Shigella* is a common intestinal pathogen, and its relative abundance is associated with pro-inflammatory states (52 [↗](#)). *Salmonella* can secrete bacterial virulence proteins, which are directly delivered to the host cell cytosol through the Type III secretion system and induce the expression of pro-inflammatory cytokines (53 [↗](#)), leading to inflammation. Overall, the structure of the colon microbiota was disrupted by *S. Typhimurium* ATCC14028 infection and increased the relative abundance of harmful bacteria, while therapeutic *B. velezensis* HBXN2020 enhanced the abundance of beneficial bacteria, reduced inflammation, and improved intestinal barrier function.

Due to the improved efficacy of treating *S. Typhimurium*-induced colitis, we further investigated the potential of *B. velezensis* HBXN2020 spores for disease prevention by evaluating pre-treatment. Similarly to the therapeutic effect of oral *B. velezensis* HBXN2020, pre-treatment with *B. velezensis* HBXN2020 spores effectively alleviated the symptoms of *S. Typhimurium*-induced colitis and reduced the expression levels of pro-inflammatory cytokines and increased anti-inflammatory cytokine levels in the colon of *S. Typhimurium*-infected mice. Furthermore, pre-treatment with *B. velezensis* HBXN2020 spores also promoted the expression levels of intestinal barrier proteins in the colon of *S. Typhimurium*-infected mice. In addition, the microbiota composition was also dramatically modulated by prophylactic *B. velezensis* HBXN2020 in mice with colitis, although the community richness and diversity revealed by alpha diversity index was not significantly impacted in mice with colitis by prophylactic-treatment. Further analysis at the genus level revealed that prophylactic *B. velezensis* HBXN2020 clearly increased SCFAs-producing bacteria including *Lactobacillus* and *Akkermansia*, which was consistent with the results of therapeutic *B. velezensis* HBXN2020. Similarly, harmful bacteria such as *Alistipes*, *Bacteroides*, *Escherichia-*

Shigella, and *Enterococcus* were significantly enriched in the PBS+STm group. *Alistipes* is a facultative - pathogenic bacteria and promotes the development of inflammation, obesity, and colorectal cancer (54 [↗](#)). Based on those results, we found that prophylactic *B. velezensis* HBXN2020 alleviates *S. Typhimurium* induced colitis by increasing the abundance of beneficial bacteria, attenuating pro-inflammatory response of the intestine, and increasing intestinal barrier function.

Conclusion

In conclusion, infection with *S. Typhimurium* ATCC14028 disrupts the intestinal mucosal barrier, and leading to intestinal inflammation. Both therapeutic and prophylactic, supplementing *B. velezensis* HBXN2020 spores alleviates *S. Typhimurium*-induced colonic injury by reducing *Salmonella* colonization, and enhancing the abundance of beneficial bacteria (especially *Lactobacillus* and *Akkermansia*) and intestinal barrier integrity, and decreasing inflammation.

Experimental procedures

Animals and bacterial strains

Female specific-pathogen-free (SPF) C57BL/6 mice aged 6 to 8 weeks were purchased from the Animal Experimental Center of Huazhong Agricultural University. All mice experiments were conducted in the standard SPF facility of Huazhong Agricultural University, with 12 h of light and 12 h of darkness at a temperature of 25°C and ad libitum access to food and water. The use of animals in this experiment was approved by the Animal Care and Ethics Committee of Huazhong Agricultural University (Ethics Approval Number: HZAUMO-2023-0089).

The strains used in this study are listed in Table S2, and all the primers used in Table S3. *Clostridium perfringens* were cultured in a fluid thioglycollate medium (FTG) (Hopebio, Qingdao, China) at 45°C. *Streptococcus suis*, *Pasteurella multocida*, and *Actinobacillus pleuropneumoniae* were cultured in tryptic soy broth (TSB) (BD Biosciences, MD, United States) medium supplemented with 5% (v/v) sheep serum (Solarbio, Beijing, China) at 37°C, while the other bacterial strains were cultured in Luria Bertani (LB) medium (Solarbio, Beijing, China) at 37°C. In addition, the Difco sporulation medium (DSM) was used for inducing the sporulation of *B. velezensis* HBXN2020 via the nutrient depletion method (55 [↗](#)).

Stain isolation, growth conditions and strain identification

In this study, we isolated 362 *Bacillus* strains from the feces of healthy black piglets that were free-ranging in a mountain village in Xianning City, Hubei Province, China, and 4 *Bacillus* strains were screened through initial antibacterial tests. Then, a strain with broad-spectrum antimicrobial activity was ultimately screened out by the antibacterial spectrum assay (Table S4), and named as *B. velezensis* HBXN2020. The strain was stored in 28% glycerol at -80°C. For culturing the bacterial cells, Luria-Bertani (LB) media or agar plates supplemented with 1.5% agar were used and incubated at 37°C.

A single bacterial colony was inoculated into 10 mL of LB medium and incubated overnight at 37°C with shaking (180 rpm). Then, the 16S rRNA sequence was amplified by the polymerase chain reaction (PCR) using forward primer 27F and reverse primer 1492R and similarity analysis with the National Center for Biotechnology Information (NCBI) alignment tool (BLAST) was used to identify the species.

Genome sequencing, annotation and analysis

The complete genome of *B. velezensis* HBXN2020 was sequenced using the Illumina HiSeq PE and PacBio RSII (SMRT) platforms (Shanghai Majorbio Bio-pharm Technology Co., Ltd.). Briefly, the short reads from the Illumina HiSeq PE were assembled into contigs using SPAdenovo (<http://soap.genomics.org.cn/>). The long reads from the PacBio RSII and the Illumina contigs were then aligned using the miniasm and Racon tools in Unicycler (version 0.4.8, <https://github.com/rrwick/Unicycler>) to generate long-read sequences. During the assembly process, sequence correction was performed using Pilon (version 1.22, <https://github.com/broadinstitute/pilon/wiki/Standard-Output>). Lastly, a complete genome with seamless chromosomes was obtained.

The coding sequences (CDs) of the *B. velezensis* HBXN2020 genome were predicted using Glimmer (version 3.02, <http://ccb.jhu.edu/software/glimmer/index.shtml>). The tRNA and rRNA genes were predicted using TRNAscan-SE (version 2.0, <http://trna.ucsc.edu/software/>) and Barrnap (version 0.8, <https://github.com/tseemann/barrnap>), respectively. The functional annotation of all CDSs was performed using various databases, including Swiss-Prot Database (https://web.expasy.org/docs/swiss-prot_guideline.html), Pfam Database (<http://pfam.xfam.org/>), EggNOG Database (<http://eggnog.embl.de/>), GO Database (<http://www.geneontology.org/>), and KEGG Database (<http://www.genome.jp/kegg/>). The circular map of the *B. velezensis* HBXN2020 genome was generated using CGView (version 2, <http://wishart.biology.ualberta.ca/cgview/>) (56).

Comparative genomic analysis

To elucidate the phylogenetic relationships from a whole-genome perspective, a phylogenetic tree based on the whole genome was constructed using the Type (Strain) Genome Server (TYGS) online (<https://ggdc.dsmz.de/>) (57). The average nucleotide identity (ANI) values of genome to genome were calculated using the JSpeciesWS online service (58), and a heatmap was then generated using TBtools (version 1.113) (59).

Growth curves of *B. velezensis* HBXN2020

The growth curve of *B. velezensis* HBXN2020 was recorded in flat-bottomed 100-well microtiter plates via detecting optical density at 600 nm (OD₆₀₀) at 1 h intervals using the automatic growth curve analyzer (Bioscreen, Helsinki, Finland).

Antimicrobial Assays

The *in vitro* antagonistic activity of *B. velezensis* HBXN2020 was tested using the agar well-diffusion method against 18 indicator strains (pathogens), which included 7 standard strains (*E. coli* ATCC 25922, *E. coli* ATCC 35150, *S. Typhimurium* ATCC 14028 (STm), *S. Typhimurium* SL1344, *S. aureus* ATCC 29213, *S. aureus* ATCC 43300, and *C. perfringens* CVCC 2030) and 11 clinical isolates (*E. coli* EC024, *S. Enteritidis* SE006, *S. aureus* S21, *C. perfringens* CP023, *C. perfringens* CP002, *S. suis* SC19, *S. suis* SS006, *P. multocida* PM002, *P. multocida* PM008, *A. pleuropneumoniae* APP015, and *A. pleuropneumoniae* APP017). All plates were cultured at 37°C for 16 h before observing the inhibition zone, and the diameter of the inhibition zone was measured using a vernier caliper. The presence of a clear zone indicated antagonistic activity.

Antibiotic susceptibility assays

Antimicrobial susceptibility testing was performed using the Kirby-Bauer (KB) disk diffusion method in accordance with the Clinical Laboratory Standards Institute (CLSI) guidelines (60). The antimicrobial agents tested were: Ampicillin, Meropenem, Piperacillin, Gentamycin, Tetracycline, Doxycycline, Minocycline, Erythromycin, Enrofloxacin, Ofloxacin, Sulfamethoxazole, Trimethoprim-sulfamethoxazole, Polymyxin B, Teicoplanin, Trimethoprim, Florfenicol, Spectinomycin, Nitrofurantoin, and Rifampicin. The diameter of the inhibition zone was measured using a vernier caliper.

***In vitro* resistance assay of *B. velezensis* HBXN2020**

B. velezensis HBXN2020 spores (100 μ L) or vegetative cells (100 μ L) were separately resuspended in 900 μ L of LB medium supplemented with different pH values (2, 3, 4, 5 or 6), bile salts (0.85% NaCl, 0.3%), simulated gastric fluid (SGF, HCl, pH 1.2) containing 10 g L⁻¹ of pepsin in 0.85% NaCl solution, or simulated intestinal fluid (SIF, NaOH, pH 6.8) containing 10 g L⁻¹ of trypsin in 0.05 M KH₂PO₄ solution, and incubated at 37 °C. A normal LB medium (pH 7.0) was used as the control. At predetermined time points, 100 μ L was taken from each sample, serially diluted 10-fold with sterile PBS (pH 7.2), and then spread onto LB agar plates. The plates were incubated overnight in a constant temperature incubator at 37°C, and the bacterial colonies were counted. The survival rate was calculated using the following formula: Survival rate = (number of bacteria in the treatment group/number of bacteria in the control group) $\times$ 100%.

One milliliter of *B. velezensis* HBXN2020 spores or vegetative cells were separately placed in water baths at different temperatures (37°C, 45°C, 55°C, 65°C, 75°C, 85°C or 95°C) for 20 min, with a 37°C water bath used as the control. The survival rate was calculated as described above.

***In vitro* bacterial competition**

To investigate whether *B. velezensis* HBXN2020 directly inhibited the growth of STm, we performed spot-on lawn and agar-well diffusion assays, as well as co-culture assays in liquid culture medium. In the spot-on lawn antimicrobial assays, we prepared double layers of agar by first pouring LB agar into the plate as the bottom layer. The top layer consisted of 10 mL TSB broth containing 0.7% agar with STm overnight culture. Then, 10 μ L of *B. velezensis* HBXN2020 overnight culture and cell-free supernatant (CFS) were respectively spotted onto TSB agar and incubated at 37°C for 12 h to measure the inhibition zone. A transparent zone of at least 1 mm around the spot was considered positive.

The antagonistic effect of *B. velezensis* HBXN2020-CFS against STm was determined using the agar-well diffusion assays. To collect *B. velezensis* HBXN2020-CFS, the culture was centrifuged at 9,000g for 15 min at 4°C, and the supernatant was filtered through a 0.22 μ m membrane filter (Millipore, USA). The STm lawn medium was prepared by mixing 10 mL of TSB broth containing 0.7% agar with STm overnight culture and then poured into a sterile plate covered with LB agar and Oxford cups, 8 mm diameter wells were prepared in the TSB agar after removing the cups. The wells were filled with 100 μ L of *B. velezensis* HBXN2020-CFS, LB medium or ampicillin (100 μ g mL⁻¹). The plates were incubated at 37°C for 14 h, and the inhibition zone was measured.

The co-culture assay was conducted by incubating *B. velezensis* HBXN2020 and *S. Typhimurium* ATCC14028 (carry pET28a (+), kanamycin resistance) (33 [DOI](#)) separately overnight at 37°C and diluting them to 10⁴ CFU/mL. Then, the two strains were mixed at different ratios (1:1, 1:10, 1:50 or 1:100) and co-cultured at 37°C with shaking (180 rpm). At predetermined time points, serial 10-fold dilutions were prepared for all samples and spread onto selective (kanamycin 50 μ g mL⁻¹) LB agar plates and cultured at 37°C for 12 h before bacterial counting. Viable colony counts ranged from 30 to 300 per plate.

Biosafety assessment

After a 7-day acclimation period (free access to water and food), mice were randomly divided into four treatment groups (n = 5): low-dose *B. velezensis* HBXN2020 spores group (L-HBXN2020 group, 10⁷ CFU/mouse), medium-dose *B. velezensis* HBXN2020 spores group (M-HBXN2020 group, 10⁸ CFU/mouse), high-dose *B. velezensis* HBXN2020 spores group (H-HBXN2020 group, 10⁹ CFU/mouse), and a control group. During the experimental period (15 days), mice were weighed and orally gavaged with their respective treatments once every two days. On day 15, all mice were euthanized, and blood, heart, liver, spleen, lung, kidney, ileum, cecum, and colon were collected. Blood samples were used for routine blood and biochemistry tests. Major organ tissues (heart,

liver, spleen, lung, and kidney) and a 5-mm distal segment of the colon were used for histopathology. The remaining colonic tissues were rapidly frozen in liquid nitrogen and stored at -80°C for cytokine and tight junction protein expression analysis.

***Salmonella* Typhimurium-induced mouse model of colitis**

After 7 days of acclimation (free access to water and food), mice were randomly divided into three treatment groups ($n = 8$): Control group, STm + PBS group, and STm + HBXN2020 group. The *S. Typhimurium*-induced colitis was performed as previously described (61), with slight modifications. On the first day of the experiment, all mice in the STm + PBS group and STm + HBXN2020 group were orally inoculated with 200 μL (5×10^7 CFU/mouse) of STm. On days 1, 3 and 5 following STm infection, each mouse in the STm + HBXN2020 group received 200 μL (1×10^8 CFU/mouse) of *B. velezensis* HBXN2020 spores via gavage administration. In contrast, the control group and STm + PBS group received 200 μL of sterile PBS by oral gavage. Fecal samples were collected daily following STm infection and resuspended in sterile PBS. The number of STm in mice feces from both the STm + PBS and STm + HBXN2020 groups was then determined by spreading a serial 10-fold dilution on selective LB agar plates containing 50 $\mu\text{g mL}^{-1}$ kanamycin. Throughout the entire experiment, the body weight, stool consistency and fecal occult blood of all mice were monitored daily. As shown in Table S5, disease activity index (DAI) was calculated by the sum of the scores from three parameters (62). On day 7 after STm infection, all mice were euthanized, their ileum, cecum and colon were collected. The length of colon was measured, and a 5-mm distal segment of the colon was fixed in 4% paraformaldehyde for further histopathology. The remaining colon was then rapidly frozen in liquid nitrogen and stored at -80°C for cytokine and tight junction protein expression analysis. Then, the number of STm in the ileum, cecum, and colon was determined by spreading serial 10-fold dilutions on selective LB agar plates.

To investigate the prophylactic efficacy of *B. velezensis* HBXN2020 in ameliorating STm-induced colitis, another independent experiment was conducted using six-week-old female C57BL/6 mice (SPF). After a seven-day acclimation period, mice were randomly assigned to three groups ($n = 8$): control group, PBS + STm group and HBXN2020 + STm group. On days 1, 3, 5 and 7, each mouse in the HBXN2020 + STm group received 200 μL (1×10^8 CFU/mouse) of *B. velezensis* HBXN2020 spores via gavage administration. Meanwhile, mice in the control group and PBS + STm group received 200 μL of sterile PBS by oral gavage. On day 7, all mice in the PBS + STm group and HBXN2020 + STm group were orally inoculated with 200 μL (5×10^7 CFU/mouse) of STm. Fecal samples were collected daily following STm infection from both groups and resuspended in sterile PBS. The number of STm in the feces of mice was then determined by spreading serial 10-fold dilutions on selective LB agar plates (50 $\mu\text{g mL}^{-1}$ kanamycin). Throughout the entire experiment, the body weight and DAI scores of all mice were monitored daily. On day 12, all mice were euthanized, and the ileum, cecum, and colon were collected. The colon length was measured, and a 5-mm distal segment of the colon was fixed in 4% formalin for sectioning and staining. The remaining colon was stored at -80°C for future analysis. Lastly, the number of STm in the ileum, cecum, and colon was determined using a selective LB agar plate.

Determination of cytokines and tight junction protein expression in colon tissue

Total RNA was extracted from colon tissues using TRIpure reagent (Aidlab, China), and cDNA was obtained using HiScript[®] III RT SuperMix for qPCR (+gDNA wiper) (Vazyme, China). RT-qPCR for each gene was performed in triplicate using qPCR SYBR Green Master Mix (Yeasen Biotechnology, Shanghai, China). The relative expression level of cytokine and tight junction protein genes was calculated using the $2^{-\Delta\Delta\text{Ct}}$ method with β -actin and GAPDH as reference genes. The primer sequences used in the RT-qPCR test are listed in Table S3.

Histopathology analysis

Colon tissue samples (0.5 cm) were fixed in 4% paraformaldehyde for 24 h, and the fixed tissues were then embedded in paraffin and sectioned. The sections were stained with hematoxylin and eosin (H&E) and observed and imaged using an optical microscope (Olympus Optical, Tokyo, Japan). The histopathological score included the degree of inflammatory infiltration, changes in crypt structures, and the presence or absence of ulceration and edema. The scoring criteria were determined as previously described ([34](#)).

16S rRNA gene sequencing and analysis

According to the manufacturer's instructions, colon microbial community genomes were extracted using E.Z-N.A.® Stool DNA Kit (Omega; D4015-01), and quality was detected by 1% agarose gel electrophoresis. The 16S rRNA V3-V4 variable region was amplified by PCR using universal primers 338F (5'-ACTCCTACGGGGGCGAG-3') and 806R (5'-GACTACHVGGGTWTCTAAT-3'). The PCR products were examined by electrophoresis on 2% agarose gels and then purified with the AxyPrep DNA gel extraction kit (Axygen Biosciences, USA). A sequencing library was constructed using the NEXTFLEX® Rapid DNA-Seq Kit, and sequencing was performed using the Illumina MiSeq platform. Raw reads were quality evaluated and filtered by fastp (version 0.20.0) and merged using FLASH (version 1.2.7). The optimized sequences were clustered into operational taxonomic units (OTUs) based on 97% sequence similarity using UPARSE (version 7.1). The representative sequences of each OTU was classified by RDP classifier (version 2.2; confidence threshold value, 0.7). Alpha diversity was assessed using the ACE, chao, Shannon, and Simpson indices. The β -diversity analysis was performed using Bray-Curtis distances and visualized through principal component analysis (PCA). Linear discriminant analysis (LDA) effect size (LefSe) was used to identify differential microbiota between groups.

Statistical Analysis

Statistical analysis was performed using GraphPad Prism 8.3.0 (GraphPad Software, San Diego, CA, USA) and Excel (Microsoft, Redmond, USA). Data are presented as mean $\pm$ standard error of the mean (SEM). Differences between two groups were evaluated using two-tailed unpaired Student's t-test, and all other comparisons were conducted using one-way analysis of variance (ANOVA) with Tukey's multiple comparisons test. For all analyses, significance differences are denoted as: *, $P < 0.05$, **, $P < 0.01$, and ***, $P < 0.001$.

Data availability

The complete genome sequence of *Bacillus velezensis* HBXN2020 has been deposited in GenBank and assigned accession numbers: CP119399.1

Conflict of interest

We declare no conflict of interest.

Acknowledgements

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Author contributions

Linkang Wang: Conceptualization, Methodology, Data curation, Formal analysis, Writing - original draft. **Haiyan Wang, Xinxin Li, Mengyuan Zhu:** Methodology, Data curation, Planned and performed the experiments. **Dongyang Gao, Dayue Hu, Zhixuan Xiong:** Methodology, Writing - Reviewing and Editing. **Xiangmin Li:** Supervision, Writing-reviewing and editing. **Ping Qian:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing-Reviewing and Editing. All authors read and approved the submitted version of the paper.

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Reviewer #1 (Public Review):

Summary:

Wang and colleagues presented an investigation of pig-origin bacteria *Bacillus velezensis* HBXN2020, for its released genome sequence, in vivo safety issue, probiotic effects in vitro, and protection against *Salmonella* infection in a murine model. Various techniques and assays are performed.

Strengths:

An extensive study on the probiotic properties of the *Bacillus velezensis* strain HBXN2020.

Weaknesses:

- The main results are all descriptive, without new insight advancing the field or a mechanistic understanding of the observed protection.
- Most of the results and analysis parts are separated without a link or any story-telling to deliver a concise message.
- For the Salmonella Typhimurium-induced mouse model of colitis, it is not clear how an oral infection of C57BL/6 would lead to colitis. Streptomycin is always pretreated (https://link.springer.com/protocol/10.1007/978-1-0716-1971-1_17).
<https://doi.org/10.7554/eLife.93423.1.sa2>

Reviewer #2 (Public Review):

Summary:

In this study, Wang and colleagues study the potential probiotic effects of *Bacillus velezensis*. *Bacillus* species have the potential benefit of serving as probiotics due to their ability to form endospores and synthesize secondary metabolites. *B. velezensis* has been shown to have probiotic effects in plants and animals but data for human use are scarce, particularly with respect to salmonella-induced colitis. In this work, the authors identify a strain of *B. velezensis* and test it for its ability to control colitis in mice.

Key findings:

- (1) The authors sequence an isolate for *B. velezensis* - HBXN2020 and describe its genome (roughly 4 mb, 46% GC-content etc).
- (2) The authors next describe the growth of this strain in broth culture and survival under acid and temperature stress. The susceptibility of HBXN2020 was tested against various antibiotics and against various pathogenic bacteria. In the case of the latter, the authors set out to determine if HBXN2020 could directly inhibit the growth of pathogenic bacteria. Convincing data, indicating that this is indeed the case, are presented.
- (3) To determine the safety profile of HBXN2020 (for possible use as a probiotic), the authors infected the strain in mice and monitored weight, together with cytokine profiles. Infected mice displayed no significant weight loss and expression of inflammatory cytokines remained unchanged. Blood cell profiles of infected mice were consistent with that of uninfected mice. No significant differences in tissues, including the colon were observed.
- (4) Next, the authors tested the ability of HBXN2020 to inhibit the growth of *Salmonella typhimurium* (STm) and demonstrate that HBXN2020 inhibits STm in a dose-dependent manner. Following this, the authors infect mice with STm to induce colitis and measure the ability of HBXN2020 to control colitis. The first outcome measure was a reduction in STm in faeces. Consistent with this, HBXN2020 reduced STm loads in the ileum, cecum, and colon. Colon length was also affected by HBXN2020 treatment. In addition, treatment with HBXN2020 reduced the appearance of colon pathological features associated with colitis, together with a reduction in inflammatory cytokines.
- (5) After noting the beneficial (and anti-inflammatory effects) of HBXN2020, the authors set out to investigate the effects on microbiota during treatment. Using a variety of algorithms, the authors demonstrate that upon HBXN2020 treatment, microbiota composition is restored to levels akin to that seen in healthy mice.

(6) Finally, the authors assessed the effect of using HBXN2020 as prophylactic treatment for colitis by first treating mice with the spores and then infecting them with STm. Their data indicate that treatment with HBXN2020 reduced colitis. A similar beneficial impact was seen with the gut microbiota.

Strengths:

- (1) Good use of in vitro and animal models to demonstrate a beneficial probiotic effect.
- (2) Most observations are supported using multiple approaches.
- (3) The mouse experiments are very convincing.

Weaknesses:

- (1) Whilst a beneficial effect is observed, there is no investigation of the mechanism that underpins this.
- (2) The mouse experiments would have benefited from the use of standard anti-inflammatory therapies to control colitis. That way the authors could compare their approach of using bacillus spores with the current gold standard for treatment.

<https://doi.org/10.7554/eLife.93423.1.sa1>

Reviewer #3 (Public Review):

Summary:

The manuscript by Wang et al. investigates the effects of *B. velezensis* HBXN2020 in alleviating *S. Typhimurium*-induced mouse colitis. The results showed that *B. velezensis* HBXN2020 could alleviate bacterial colitis by enhancing intestinal homeostasis (decreasing harmful bacteria and enhancing the abundance of *Lactobacillus* and *Akkermansia*) and gut barrier integrity and reducing inflammation. Overall, the manuscript is of potential interest to readers.

Strengths:

B. velezensis HBXN2020 is a novel species of *Bacillus* that can produce a great variety of secondary metabolites and exhibit high antibacterial activity against several pathogens. *B. velezensis* HBXN2020 is able to form endospores and has strong anti-stress capabilities. *B. velezensis* HBXN2020 has a synergistic effect with other beneficial microorganisms, which can improve intestinal homeostasis.

Weaknesses:

There are few studies about the clinical application of *Bacillus velezensis*. Thus, more studies are still needed to explore the effectiveness of *Bacillus velezensis* before clinical application.

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